IF THE ASSOCIATED POINTS LIE IN THE IMAGE OF A FLAT MORPHISM THEN THE CANONICAL MAP TO THE PUSHFORWARD IS UNIVERSALLY INJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a reduction to weakly associated points of the kernel and a point by point argument using faithful flatness of the local ring map along a flat morphism with preimages of all associated points

Editorial note. The label lemma-universally-separating names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Flatness and weak associated points, and the criterion that a module is zero when it has no weakly associated point
- The inclusion of the weakly associated points of a subquotient into those of the ambient module, and the containment of the weakly associated points of a flat finite type module in the associated points of the fibres
- Faithful flatness of the map of local rings induced by a flat morphism at a point of the image, and the resulting injectivity of the base changed stalk

Source reference. Original lemma 05CK: sources/stacks-chow/tags.txt, source lines 6923–6923. Prerequisite pointer invoked at proof line(s) 1208; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 05IJ: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 733–733. Prerequisite pointer invoked at proof line(s) 1193; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0CW2: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 1031–1031. Prerequisite pointer invoked at proof line(s) 1180; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0CTZ: sources/stacks-chow/tags.txt, source lines 16084–16084. Prerequisite pointer invoked at proof line(s) 1191; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0CU0: sources/stacks-chow/tags.txt, source lines 16085–16085. Prerequisite pointer invoked at proof line(s) 1188; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let Y be an algebraic space over a scheme S. Let $g : X' \to X$ be a flat morphism of algebraic spaces over Y with X locally of finite type over Y. Let $\mathcal{F}$ be a finite type quasi-coherent $\mathcal{O}_X$-module which is flat over Y. If $Ass_{X/Y}(\mathcal{F}) \subset g(|X'|)$ then the canonical map*

$$\mathcal{F} \longrightarrow g_*g^*\mathcal{F}$$

is injective, and remains injective after any base change.

Proof. The final assertion means that $\mathcal{F}_Z \to (g_Z)_*g_Z^*\mathcal{F}_Z$ is injective for any morphism $Z \to Y$. Since the assumption on the relative assassin is preserved by base change (Lemma 0CW2) it suffices to prove the injectivity of the displayed arrow.

Let $\mathcal{K} = \mathrm{Ker}(\mathcal{F} \to g_*g^*\mathcal{F})$. Our goal is to prove that $\mathcal{K} = 0$. In order to do this it suffices to prove that $\mathrm{WeakAss}_X(\mathcal{K}) = \emptyset$, see Divisors on Spaces, Lemma 0CU0. We have $\mathrm{WeakAss}_X(\mathcal{K}) \subset \mathrm{WeakAss}_X(\mathcal{F})$, see Divisors on Spaces, Lemma 0CTZ. As $\mathcal{F}$ is flat we see from Lemma 05IJ that $\mathrm{WeakAss}_X(\mathcal{F}) \subset \mathrm{Ass}_{X/Y}(\mathcal{F})$. By assumption any point x of $\mathrm{Ass}_{X/Y}(\mathcal{F})$ is the image of some $x' \in |X'|$. Since g is flat the local ring map $\mathcal{O}_{X,\overline{x}} \to \mathcal{O}_{X',\overline{x}'}$ is faithfully flat, hence the map

$$\mathcal{F}_{\overline{x}} \longrightarrow (g^*\mathcal{F})_{\overline{x}'} = \mathcal{F}_{\overline{x}} \otimes_{\mathcal{O}_{X,\overline{x}}} \mathcal{O}_{X',\overline{x}'}$$

is injective (see Algebra, Lemma 05CK). Since the displayed arrow factors through $\mathcal{F}_{\overline{x}} \to (g_*g^*\mathcal{F})_{\overline{x}}$, we conclude that $\mathcal{K}_{\overline{x}} = 0$. Hence x cannot be a weakly associated point of $\mathcal{K}$ and we win. □